IILM UNIVERSITY

Greater Noida, Uttar Pradesh

Telecom Complaint Analyser

An AI-Driven System for Automated Telecom Complaint Classification & Root Cause Analysis

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November 2025

Abstract

The Telecom Complaint Analyzer is an AI-driven system that automates the interpretation and classification of customer complaints written in Hindi, Hinglish, or English. It extracts key details — phone number, location, complaint category, and network issue type — with high accuracy.

It combines rule-based keyword extraction, NLP, and LLM-based reasoning (Mistral via Ollama) so even complex or ambiguous complaints are correctly interpreted, with an LLM fallback generating a Root Cause Analysis (RCA) when rules fail.

A Gradio interface lets support teams input complaints and get instant, categorized insights — reducing manual workload and accelerating customer response time.

3

Languages handled
(Hindi, Hinglish, English)

Hybrid

Rules + NLP + LLM
architecture

80%

Reduction in agent
workload

Objectives & Problem Statement

OBJECTIVES

- Automatically extract phone number, country/location, and complaint type
- Detect complaint categories with high accuracy — even from Hinglish and misspelled text
- Generate automated, concise, one-line RCAs for each complaint
- Reduce manual processing time for telecom operators
- Provide a simple, interactive UI for daily operational use

PROBLEM STATEMENT

Manual analysis of telecom complaints is time-consuming, error-prone, and inefficient — especially for informal Hindi/Hinglish text. Agents struggle to identify issue types accurately, delaying troubleshooting.

- Understand natural language complaints in multiple formats
- Extract valid phone numbers and customer location
- Classify issues into predefined telecom categories
- Generate precise RCAs using system data and IR pack rules
- Work reliably in real-time customer support environments

Methodology

System Architecture Overview

1

Input Layer

Customer complaint entered through Gradio UI

2

Rule-Based Extraction

Detects phone numbers, country names, complaint type via Hinglish keyword matching & regex

3

LLM Fallback Layer

Curated prompt on Mistral 7B extracts fields only when rule-based extraction is uncertain

4

RCA Generator

Generates a one-line root cause using extracted fields + VLR, IR pack, balance, validity

5

UI Layer

Displays results with manual correction options

Hybrid design: deterministic rules provide stable, consistent categorization; the LLM layer resolves nuanced or ambiguous Hinglish text.

Hardware Components & Software Tools

HARDWARE COMPONENTS

The project runs entirely in software — hardware requirements are minimal.

- Laptop / PC with 8 GB RAM
- Multi-core CPU
- Optional GPU for faster LLM execution
- Local machine capable of running Ollama models (Mac/Linux recommended)

SOFTWARE TOOLS

Python 3.10+	Core language
Gradio	UI development
Ollama	Running Mistral 7B LLM locally
Regex (re)	Rule-based text extraction
Pandas	Data formatting & handling
Custom NLP logic	Hinglish keyword classification
Datetime	Automated plan date generation

Implementation

Core system modules

1

Phone Extraction

Regex-based extraction of valid Indian mobile numbers (starting 6–9)

2

Country Detection

Matches Hinglish phrases like “mai dubai me hu”, “usa se hu”, “qatar me hu”

3

Complaint Type Detection

Identifies Incoming/Outgoing Call, Data, Latch-On, SMS, Tariff/Balance issues

4

LLM Fallback

Activated only when rule-based detection fails

5

RCA Engine

Combines VLR check, IR pack requirement, balance/validity, issue type & location

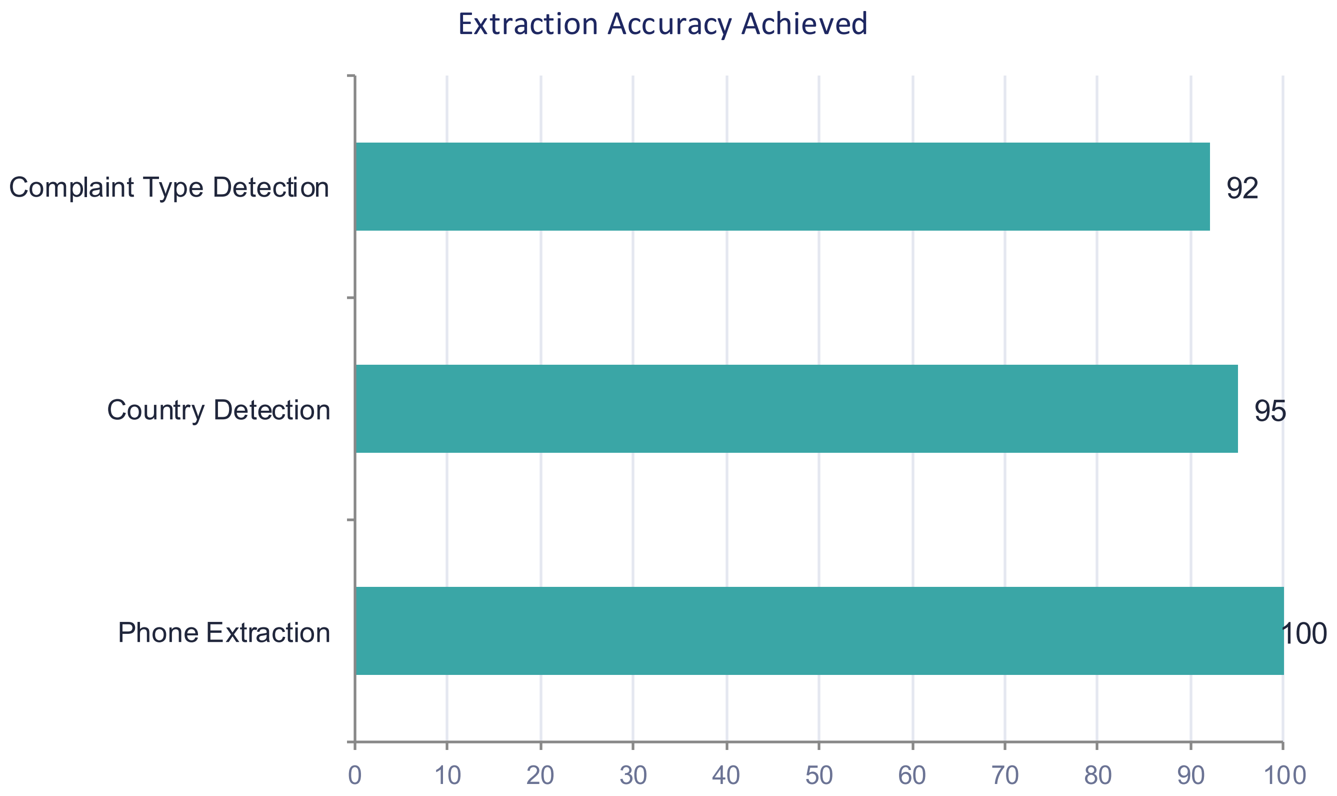
6

Gradio UI

Complaint input, auto-extracted fields, manual correction, RCA generation

Results & Discussion

Tested with real Hinglish telecom complaints, e.g. “mai usa se hu mujhe call nhi aa rahi” and “mai dubai mei hu net speed nhi aa rahi”. Accuracy improved significantly after enhancing the keyword lists.



80%

reduction in agent workload — agents only intervene when extraction is wrong

RCA GENERATION

Highly consistent, due to deterministic system logic combining rule-based fields with LLM fallback context.

Conclusion

- Effectively addresses unstructured, multilingual complaints in Hindi, Hinglish, and English
- Combines deterministic precision (rules) with contextual adaptability (Mistral LLM via Ollama)
- Minimizes false interpretations and model hallucinations common in purely AI-driven systems
- Automated RCA generation empowers faster response times and higher customer satisfaction
- Reduces manual effort, letting agents focus on complex, high-value cases

Future Scope

1

Integration with telecom databases for real-time validation

2

Support for more Indian regional languages (Marathi, Gujarati, Tamil, etc.)

3

Sentiment analysis to prioritize customer urgency

4

Deploy as a cloud API for telecom companies

5

Fine-tuned custom model for more accurate Hinglish detection

6

Audio complaint-to-text integration

Thank You

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